THE ZEROTH COHOMOLOGY OF THE DERIVED INTERNAL HOM COMPUTES HOM OVER AN OBJECT AND OVER THE WHOLE SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the choice of a K-injective and a K-flat representative, the K-injectivity of the internal hom complex, and the computation by sections.

Editorial note. The label lemma-section-RHom-over-U names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- K-injective and K-flat representatives. Same chapter, the lemma that the internal hom from a K-flat complex into a K-injective complex is K-injective. Supplies the K-injectivity of the internal hom complex used to compute its cohomology.
- The H^0 of an internal hom into a K-injective complex as derived Hom. Same chapter, the lemma identifying the zeroth cohomology of the internal hom complex with the derived Hom groups globally and over an object. Supplies the final identification in the proof.

Source reference. Original reference 0A8T: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8789–8789. Prerequisite pointer invoked at proof line(s) 8890; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A8R: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8720–8720. Prerequisite pointer invoked at proof line(s) 8892; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let L, M be objects of $D(\mathcal{O})$. For every object U of $\mathcal{C}$ we have*

$$H^0(U, R\mathcal{H}om(L, M)) = \mathrm{Hom}_{D(\mathcal{O}_U)}(L|_U, M|_U)$$

and we have $H^0(\mathcal{C}, R\mathcal{H}om(L, M)) = \mathrm{Hom}_{D(\mathcal{O})}(L, M)$.

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Proof. Choose a K-injective complex $\mathcal{I}^\bullet$ of $\mathcal{O}$-modules representing M and a K-flat complex $\mathcal{L}^\bullet$ representing L. Then $\mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet)$ is K-injective by Lemma 0A8T. Hence we can compute cohomology over U by simply taking sections over U and the result follows from Lemma 0A8R. □